

Ali Nawaf

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Education

Case Western Reserve University | Cleveland, OH

- Bachelors of Science in Computer Science, Secondary Major: Mathematics. May 2027
- Masters of Science in Machine Learning and Artificial Intelligence. May 2027

Skills

Languages: C, CUDA, Python, Java, Swift, SQL, HTML, CSS, JavaScript (Node.js, Next.js), Rust.

Machine Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Keras, VAEs, CNN, RESNET, YOLO, LLMS, BERT.

Cloud & DevOps: Azure, Power BI, SLURM, HPC, Kubernetes, AWS, Lamda functions, Linux, Jenkins, REST APIs.

Experience

Machine Learning Research Assistant | Huston Methodist Hospital | Remote Aug 2025 - Present

- Independently researched and developed a novel hypothesis for cardiovascular risk prediction, applying a pioneer ML model to test the theory against a real-world CT scan dataset, culminating in **90%** predictive accuracy.
- Engineered a signal processing pipeline to digitize raw ECG images, implementing **open-source** models and post processing algorithms to extract meaningful signals and enhance fidelity for downstream ML analysis.

Research Assistant | Case Western Reserve University | Cleveland OH | [link](#) Aug 2025 - Present

- Working with Prof. **Erman Ayday** on a **\$1.2M NIH**-funded project developing a privacy-preserving sandbox for federated genomic data sharing and ML-based **GWAS analysis**.
- Implemented (sample relatedness, population stratification) to enhance reliability for genetic association models.

Software Engineer | Eaton | Cleveland, OH Jan 2025 – May 2025

- Built a fully automated pipeline processing of **100K+ Salesforce** records per month hosted completely on **Azure**.
- Implemented batch analytics workflows, cutting latency by **35%** and lowering operational costs.
- Delivered business insights using **Power BI**, enabling data-driven decision-making for the leadership team.
- Led the team to present project results, metrics and documentation to supervisors and non-technical leadership.

Machine Learning Intern | Heads-up Hockey | Waterford, VA Dec 2024 – Aug 2025

- Automated computer vision annotation pipeline, reducing time by **70%** using **SAM2** and a **YOLO model**.
- Built an interactive **Swift UI** educational game called “[Heads up Crush](#)” with on-device **AI inference**. Which increased userbase by **100%** upon release.
- Collaborated with a team of 3 engineers to develop the company’s pioneer Hockey stick-handling **AI model** which improved accuracy in unseen testing by **30%**.
- Engineered an **end-to-end AI agent** by implementing data **pre-processing** and **post-processing** to extract goal counts, speed, and trajectory, achieving **70%** accuracy in shot trajectory prediction using the Iphone’s camera.

Projects

Medical Smart Labeler | PyTorch, Python, API, git, GitHub, Docker, Networking | [link](#)

- Reduced manual image annotation time by **60%** by developing a Python-based tool that integrated with the Label-Studio API to pre-label images using **SAM2** model.

Variational Autoencoder (VAE) Implementation | Python, PyTorch, TensorFlow, NumPy, MNIST, Matplotlib | [link](#)

- Designed and implemented a **generative deep learning model** to reconstruct images and generate new synthetic data samples from a learned **latent space**. Visualized the learned latent walk in an interactive user interface.

Network Router Simulator | C, Binary I/O, Parsing, TCP, UDP, Data Serialization | [visit](#)

- Developed a flow analyzer (hash-based NetFlow, TCP RTT), a router simulator, and an IPv4 binary trace parser in C to quantitatively analyze complex network packet data in short time.